2004 מועד ב

1. $U_{tt} + 2hU_t = c^2 U_{xx} \quad h>0$

$U(x=0,t) = U(x=L,t) = 0$

$U(x,0) = f(x)$

$\left.\frac{\partial U}{\partial t}\right|_{t=0} = \phi(x)$

נפתח פתרון בהפרדת משתנים :

$U = X(x)\,T(t)$

$$\Rightarrow \quad \ddot{T}X + 2h\dot{T}X = c^2 T X''$$

$$\frac{\ddot{T}}{c^2 T} + \frac{2h\dot{T}}{c^2 T} = \frac{X''}{X} = -a^2$$

$$\Rightarrow \quad X(x) = A_n \sin\left(\frac{n\pi x}{L}\right) \qquad \text{כרגיל}$$

$n=1,2,3$

$$\Rightarrow \quad \frac{\ddot{T}}{c^2 T} + \frac{2h}{c^2}\frac{\dot{T}}{T} = -a^2 = -\frac{\pi^2 n^2}{L^2}$$

$$\Rightarrow \quad \ddot{T} + 2h\dot{T} = -\frac{\pi^2 n^2}{L^2}c^2 T$$

$$\Rightarrow \quad \ddot{T} + 2h\dot{T} + \frac{\pi^2 n^2 c^2}{L^2}T = 0$$

נניח $T = e^{r}$

נציב במשוואה ונקבל

$$r^2 + 2hr + \frac{\pi^2 n^2 c^2}{L^2} = 0$$

$$\Rightarrow r_{1,2} = \frac{-2h \pm \sqrt{4h^2 - 4\frac{\pi^2 n^2 c^2}{L^2}}}{2}$$

נניח $\lambda_1 > h$

$$= -h \pm \sqrt{h^2 - \lambda_n^2}$$

$$= -h \pm i\sqrt{\lambda_n^2 - h^2}$$

לכן הפתרון הזמני

$$T_n(t) = A_n e^{-ht} \cos(\sqrt{\lambda_n^2 - h^2}\, t)$$

$$+ B_n e^{-ht} \sin(\sqrt{\lambda_n^2 - h^2}\, t)$$

הפתרון הכללי

$$u(x,t) = \sum_{n=1}^{\infty} e^{-ht} \sin\left(\frac{n\pi x}{L}\right)\Big[A_n \cos\sqrt{\lambda_n^2 - h^2} + +$$

$$B_n \sin(\sqrt{\lambda_n^2 - h^2}\, t)\Big]$$

בעזרת תנאי ההתחלה

$$f(x) = u(x,0) = \sum_{n=1}^{\infty} \left(\sin\left(\frac{n\pi x}{L}\right)\right) A_n$$

$$\Rightarrow \quad A_n = \frac{2}{L} \int_0^L \sin\left(\frac{n\pi x}{L}\right) f(x)\, dx$$

$$\phi(x) = \left.\frac{\partial u}{\partial t}\right|_{t=0} = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{L}\right)\Big[-hA_n + \sqrt{\lambda_n^2 - h^2}\, B_n\Big]$$

ב,

$\Delta U = 0$

$a < b$

$U(r=a,\theta) = 0$

$U(r=b) = Ay^2 = Ab^2 \sin^2\theta = A\frac{b^2}{2}[1-\cos 2\theta]$

הפתרון הכללי באיזור זה

$$U(r,\theta) = \sum_{n=1}^{\infty} (a_n r^n + b_n r^{-n})\cos(n\theta)$$
$$+ (c_n r^n + d_n r^{-n})\sin(n\theta)$$
$$+ a_0 + b_0 \ln r$$

נשתמש בתנאי השפה ב $r=a$

$$0 = U(a,\theta) = \sum_{n=1}^{\infty} (a_n a^n + b_n a^{-n})\cos(n\theta)$$
$$+ (c_n a^n + d_n a^{-n})\sin(n\theta)$$
$$+ a_0 + b_0 \ln a$$

$$\Rightarrow \quad \left.\begin{aligned} a_n a^n + b_n a^{-n} &= 0 \\ c_n a^n + d_n a^{-n} &= 0 \end{aligned}\right\} \forall n$$
$$a_0 + b_0 \ln a = 0$$

ב- $r=b$:

$$\frac{Ab^2}{2}(1-\cos 2\theta)\ U(b,\theta) = \sum_{n=1}^{\infty} (a_n b^n + b_n b^{-n})\cos(n\theta)$$
$$+ (c_n b^n + d_n b^{-n})\sin(n\theta)$$
$$+ a_0 + b_0 \ln b$$

$\forall n \quad c_n b^n + d_n b^{-n} = 0$

$\Rightarrow \quad n=2 \quad a_2 b^2 + b_2 b^{-2} = -\frac{Ab^2}{2}$

$n \neq 2 \quad a_n b^n + b_n b^{-n} = 0$

$\frac{Ab^2}{2} = a_0 + b_0 \ln b$

$n\ \forall \quad c_n = d_n = 0$ [illegible]

$n \neq 2 \quad a_n = b_n = 0$

$$a_0 + b_0 \ln a = 0$$

$$a_0 + b_0 \ln b = \frac{Ab^2}{2}$$

$$\Rightarrow \quad a_0 = \frac{-\frac{Ab^2}{2}\ln a}{\ln b - \ln a} = \frac{Ab^2}{2}\frac{\ln a}{\ln\left(\frac{a}{b}\right)}$$

$$b_0 = \frac{\frac{Ab^2}{2}}{\ln\left(\frac{b}{a}\right)} = -\frac{\frac{Ab^2}{2}}{\ln\left(\frac{a}{b}\right)}$$

$$a_2 a^2 + b_2 a^{-2} = 0$$

$$a_2 b^2 + b_2 b^{-2} = -\frac{Ab^2}{2}$$

$$\Rightarrow \quad a_2 = \frac{\frac{Ab^2 a^{-2}}{2}}{a^2 b^{-2} - a^{-2} b^2} = \frac{Ab^4}{2(a^4 - b^4)}$$

$$b_2 = \frac{-\frac{Ab^2 a^2}{2}}{a^2 b^{-2} - a^{-2} b^2} = \frac{-Ab^4 a^4}{2(a^4 - b^4)}$$

סה"כ

$$U(r,\theta) = \frac{Ab^2}{2}\frac{1}{\ln\left(\frac{a}{b}\right)}\left(\ln\left(\frac{a}{r}\right)\right) + \Big[\frac{A}{\quad\quad}$$

$$+ \frac{Ab^4}{2(a^4 - b^4)}\left[r^2 - a^4 r^{-2}\right]\cos(2\theta)$$

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$u(x=0,t)=u(x=+\infty,t)=0$

$0<x<+\infty$

$u(x,0)=f(x)$

$f(0)=0$

נבצע המשך אנטי-סימטרי לפתרון כך ש:

$$f(-x)=-f(x) \qquad x>0$$

ואז מתקיים *

$$u(x,t)=\frac{1}{\sqrt{4\pi kt}}\int_{-\infty}^{\infty} f(\zeta)e^{-\frac{(x-\zeta)^2}{4kt}}d\zeta=$$

$$=\frac{1}{\sqrt{4\pi kt}}\left[\int_{0}^{\infty} f(\zeta)e^{-\frac{(x-\zeta)^2}{4kt}}d\zeta+\int_{-\infty}^{0} f(\eta)e^{-\frac{(x-\eta)^2}{4k}}d\eta\right]$$

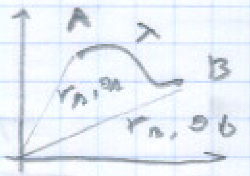
$$\eta\to-\zeta \qquad \int_{-\infty}^{0}\to\int_{\infty}^{0} \qquad f(-\eta)=-f(\zeta)$$

$$d\zeta\to-d\eta$$

$$\to\frac{1}{\sqrt{4\pi kt}}\left[\int_{0}^{\infty} f(\zeta)\left[e^{-\frac{(x-\zeta)^2}{4kt}}+e^{-\frac{(x+\zeta)^2}{4kt}}\right]d\zeta\right]$$

$\frac{q}{r}$

$\phi = \int \frac{dq}{r} dr$



$$I[r(\theta)] = \int_A^B \frac{\lambda}{r} dl = \int_{\theta_A}^{\theta_B} \frac{\lambda}{r} \sqrt{r^2 + r'^2}\, d\theta$$

$$dl^2 = dx^2 + dy^2 = r^2 d\theta^2 + dr^2 = d\theta^2 (r^2 + r'^2)$$

$x = r\cos\theta \qquad dx = \cos\theta\, dr + r\sin\theta\, d\theta$

$dy = \sin\theta\, dr + r\cos\theta\, d\theta$

$$= \int_{\theta_A}^{\theta_B} \lambda \sqrt{1 + \left(\frac{r'}{r}\right)^2}\, d\theta$$

$$F(r, r', \cancel{\theta}) = F(r, r')$$

$2\left(\frac{r'}{r}\right) \cdot \frac{1}{r}$

$$r' \frac{\partial F}{\partial r'} - F = \text{const} \qquad \Leftarrow$$

$$\text{const} = r' \frac{1}{\cancel{2}\sqrt{1 + \left(\frac{r'}{r}\right)^2}} \cdot \frac{\cancel{2} r'}{r^2} - \sqrt{1 + \left(\frac{r'}{r}\right)^2} =$$

$$= \frac{r'^2 / r^2}{\sqrt{1 + \left(\frac{r'}{r}\right)^2}} - \sqrt{1 + \left(\frac{r'}{r}\right)^2} =$$

$$= \frac{\frac{r'^2}{r^2} - 1 - \frac{r'^2}{r^2}}{\sqrt{1 + \left(\frac{r'}{r}\right)^2}} = \frac{-1}{\sqrt{1 + \left(\frac{r'}{r}\right)^2}}$$

$$\Rightarrow 1 + \left(\frac{r'}{r}\right)^2 = C$$

$$\frac{r'}{r} = C$$

$$\frac{dr}{r} = C\, d\theta$$

$$\Rightarrow \ln(r) = A\theta + B$$

$$r(\theta) = e^{A\theta + B} \qquad r_A, \theta_A = 0 \qquad r_B, \theta_B = \theta$$

ח'))

$$r(\theta) = Ae^{B\theta}$$

$$r(0) = r_A \qquad A = r_A$$

$$r(\theta_B) = r_B = r_A e^{B\theta_B}$$

$$\ln\left(\frac{r_B}{r_A}\right) = B\theta_B$$

$$B = \ln\left(\frac{r_B}{r_A}\right)\cdot\frac{1}{\theta_B}$$

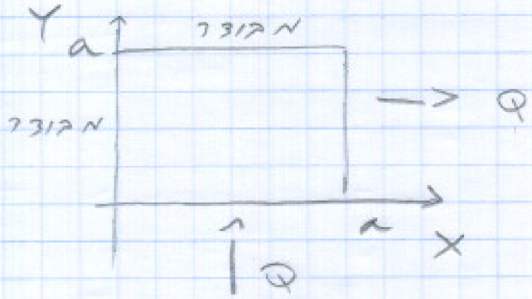
$$r(\theta) = r_A e^{\ln\left(\frac{r_B}{r_A}\right)^{\frac{1}{\theta_B}}\cdot\theta}$$

$$= r_A e^{\frac{\theta}{\theta_B}\ln\left(\frac{r_B}{r_A}\right)}$$

5.

K

$\Delta T = 0$



$$\vec{J} = -K\vec{\nabla}T$$

$$\left.\frac{\partial T}{\partial x}\right|_{x=0} = 0$$

$$\left.\frac{\partial T}{\partial y}\right|_{y=a} = 0$$

$$Q(+\hat{y}) = -K\left.\frac{\partial T}{\partial y}\right|_{y=0}\hat{y}$$

$$\left.\frac{\partial T}{\partial y}\right|_{y=0} = -\frac{Q}{K}$$

$$Q(+\hat{x}) = -K\left.\frac{\partial T}{\partial x}\right|_{x=a}\hat{x}$$

$$\left.\frac{\partial T}{\partial x}\right|_{x=a} = -\frac{Q}{K}$$

נחפש פתרון פרטי המקיים $\Delta T_p = 0$ ואז תנאי

הכי כללי: נחפש $T_p = Ax^2 + By^2 + Cx + Dy\,y + E$

$$\Rightarrow \quad 0 = \Delta T_p = 2A + 2B = 0 \qquad B = -A$$

$$T_p = A(x^2 - y^2) + Bx + Cy + D$$

$$\left.\frac{\partial T}{\partial x}\right|_{x=0} = 0 \qquad 2A\cdot 0 + B = 0$$

$$B = 0$$

$$\left.\frac{\partial T}{\partial y}\right|_{y=a} = -2Aa + C = 0$$

$$2Aa = C$$

$$T_p = A(x^2 - y^2) + 2Aay + B$$

$$-\frac{Q}{K} = \left.\frac{\partial T}{\partial x}\right|_{x=a} = 2Aa \cancel{+2Aa}$$

$$A = -\frac{Q}{2Ka}$$

$$-\frac{Q}{K} = \left.\frac{\partial T}{\partial y}\right|_{y=0} = -2\cancel{A}a + 2Aa$$

$$T_p = -\frac{Q}{2Ka}(x^2 - y^2) - \frac{Q}{K}y + B$$